LAW-E

Thermodynamic Governance Framework

Runtime Stability and Governance for Artificial Intelligence Systems

Technical White Paper — Version 3.1

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One stability law, many domains of application

1. The Hidden Cost of Unobservable AI

Modern AI systems are deployed with a known limitation: they can produce incorrect, unstable, or unpredictable outputs.

Today, this limitation is managed through disclaimers. However, disclaimers do not reduce risk — they only transfer it.

For organizations operating AI systems in production, this creates three critical challenges:

- Unpredictable operational costs
- Unquantified reliability risk
- Increasing regulatory exposure

Despite advances in model performance, there is currently no standard way to measure instability at runtime.

1.1 Introduction

Artificial intelligence systems are rapidly increasing in scale, autonomy, and operational complexity. Large language models, autonomous agents, and multi-model pipelines are now deployed continuously in production environments where stability, reliability, and operational efficiency must be maintained over time.

Instabilities in AI systems may manifest through:

- Hallucinated or unreliable outputs
- Cascading informational errors
- Unstable reasoning chains
- Uncontrolled computational cost

- Degradation of performance over time

Most existing governance approaches are reactive. The Law-E framework introduces a different paradigm:

> *AI systems are treated as informational dynamical systems whose stability can be observed at runtime.*

2. From Disclaimer to Measurable Stability

Current artificial intelligence systems are routinely deployed with disclaimers stating that outputs may be incorrect, incomplete, or unreliable.

These disclaimers are not merely precautionary. They reflect a fundamental limitation: the absence of a measurable, runtime indicator of system stability.

In practice, this means that:

- Errors cannot be anticipated in real time
- Instability cannot be quantified
- Reliability cannot be continuously evaluated

As a result, risk is not managed — it is deferred.

The Law-E framework introduces a different paradigm. Instead of treating error as an unavoidable and unquantified phenomenon, it establishes a measurable signal associated with system instability during execution.

In this context, uncertainty is not eliminated — but becomes:

- Measurable
- Trackable
- Actionable

> *This shift marks a transition: from static disclaimers to dynamic stability observability. Reliability is no longer assumed — it is continuously evaluated.*

2.1 Scientific Foundations

The theoretical foundations are described in: Thermodynamic Governance of Artificial Intelligence Systems — Foundational Framework (Zenodo).

Reference: https://doi.org/10.5281/zenodo.19031860

This document focuses on the operational layer.

3. AI Systems as Informational Dynamical Systems

AI systems evolve as stateful dynamical systems.

Stable systems:	maintain coherent information flows and converge toward consistent outputs
Unstable systems:	accumulate noise, diverge in reasoning, and degrade over time

The framework introduces G(x), a latent thermodynamic stability variable, operationally approximated by the Energy Stability Index ESI(x).

G(x) — The unified governance variable

System stability is ultimately expressed through G(x), a unified governance variable integrating instantaneous state, accumulated instability, and runtime conditions. G(x) is the signature metric of the Law-E framework — the single variable from which all governance decisions are derived.

4. Thermodynamic Interpretation

AI systems transform information continuously. From a thermodynamic perspective:

- Stable systems minimize informational entropy
- Unstable systems increase dissipation and noise

This enables the use of dynamical systems theory, Lyapunov-like stability analysis, and energy-based modeling — applied to runtime AI behavior.

5. Cross-Domain Normalization

AI signals are heterogeneous — spanning semantic outputs, agent decisions, and infrastructure metrics. The Normalizer converts these heterogeneous signals into a unified thermodynamic flow, enabling cross-domain stability measurement.

6. Energy Stability Index (ESI)

$$ESI = 100 \times (\beta_1 S + \beta_2 P + \beta_3 Q + \beta_4 C)$$

Where:

- S = historical stability

- P = energy efficiency
- Q = measurement quality
- C = computational reliability

The coefficients β_1–β_4 are calibrated based on domain-specific constraints and can be dynamically adjusted depending on deployment context. The index yields a normalized score in [0, 100], where higher values indicate greater stability.

7. Metric Layer

The Law-E framework operates through a set of runtime metrics capturing system behavior across multiple dimensions:

- Energy-related indicators (resource usage proxies)
- Stability indicators (system consistency signals)
- Performance-related measures (latency, response characteristics)

The full metric layer is designed to support real-time monitoring, governance decisions, and post-execution auditability. The metric layer is progressively exposed through the Law-E runtime environment.

8. Stability Function

The stability function is defined as:

$$V(x) = 100 - ESI(x)$$

The system is considered stable when:

$$dV/dt \leq 0$$

This Lyapunov-like condition ensures that the framework can formally characterize convergence and divergence of AI system behavior over time.

The relationship between V(x) and G(x) forms the mathematical core of the framework: V measures local instability at a given state, while G integrates this signal across time and system context into a unified governance indicator.

G(x) can be interpreted as a time-integrated transformation of V(x) under system constraints — V is local, G is global.

9. Governance Architecture

The full pipeline operates as follows:

Runtime Signals → Normalizer → Thermodynamic Flow → Aggregation → Stability Computation → Governance Decision

This pipeline converges toward a unified governance variable integrating instantaneous state, accumulated instability, and runtime conditions — from which all governance decisions are derived.

All governance decisions are ultimately derived from G(x), making it the central control variable of the system.

OBS Mode — Observation

Monitoring only. Runtime metrics are measured and logged without active intervention. Enables auditability and post-hoc analysis.

GOV Mode — Governance

Active regulation. The framework generates governance decisions (authorize / modulate / block) based on computed stability indicators.

10. Thermodynamic Aggregation

The framework introduces the concept of thermodynamic operational debt — accumulated instability over time, defined as:

$$D(t) = D(t-1) + E(t) - f(D(t-1), E(t))$$

Where f is a dissipation function. This memory mechanism enables the detection of slow-building instability patterns that would be invisible to point-in-time measurements.

11. Reference Implementation

A sandbox implementation is publicly available:

https://github.com/neomundi-tech/neomundi-sandbox

The sandbox implementation provides a practical illustration of the OBS mode, where runtime signals are captured and exposed without intervention. It demonstrates how execution-level metrics are normalized into thermodynamic flows and made available through a structured runtime interface.

It serves as a first operational entry point for evaluating runtime stability signals in real conditions.

12. Experimental Results

Objective: detect instability signals correlated with incorrect or misleading outputs — without access to ground truth labels at inference time.

TruthfulQA benchmark — ~790 questions	
Traditional systems	✗ No runtime instability signal ✗ Failures detected after impact
~79% **Law-E**	✓ Instability signals detected at runtime — before impact

Interpretation

The framework does not eliminate errors, but introduces a measurable signal associated with instability. This enables runtime monitoring, post-hoc analysis, and governance actions — without modifying the underlying model.

13. Applications

- LLM hallucination monitoring
- AI agent stability scoring
- Multi-agent observability
- Cost optimization
- Infrastructure monitoring

13.1 Operational Impact

In production environments, undetected instability can lead to incorrect automated decisions, increased support and remediation costs, and compliance and liability risks.

By introducing runtime stability observability, Law-E enables:

- Earlier detection of problematic outputs
- Reduction of costly failure cascades
- Improved auditability for regulated environments

13.2 Alignment with EU AI Act

The European AI Act introduces requirements for risk management, transparency, traceability, and monitoring of AI systems. High-risk systems must ensure continuous performance monitoring, logging, and auditability.

The Law-E framework directly aligns with these requirements by:

- Providing runtime metrics
- Enabling traceability via structured logs
- Supporting post-deployment monitoring

As such, it contributes to compliance strategies for AI systems subject to regulatory oversight.

14. Roadmap

Short-term

- Deployment of Governance Mode (GOV)
- Progressive exposure of runtime metrics
- Expansion of API observability capabilities

Mid-term

- Cross-domain integration: AI agents, infrastructure systems, distributed environments
- Structuring of the Law-E framework as a unified thermodynamic operating layer
- Extension of IP framework across measurement, aggregation, regulation, and coordination

Long-term

- Integration of thermodynamic metrics into hardware-level architectures
- Development of energy-aware AI systems
- Exploration of native thermodynamic computation paradigms

15. Intellectual Property

The Law-E framework is supported by a structured IP portfolio covering:

1. Thermodynamic measurement systems	Quantification of informational and energetic properties
2. Temporal aggregation mechanisms	Accumulation and long-term tracking of thermodynamic signals
3. Stability-constrained training architectures	Controlled exploration and convergence via stability functions
4. Distributed thermodynamic coordination	Agent interaction based on thermodynamic indicators
5. Unified governance layer	Integrated measurement, memory, regulation and decision

These inventions collectively define a new class of runtime AI governance systems. Certain implementation details remain undisclosed.

16. Conclusion

The Law-E framework introduces a thermodynamic perspective on AI system behavior, a runtime observability layer, and a path toward active governance.

AI systems have historically relied on disclaimers to manage uncertainty. Law-E replaces this paradigm by making instability measurable at runtime. Reliability is not assumed — it is continuously evaluated.

> *This approach defines a new class of runtime governance systems for AI — one where stability is not assumed, but measured, tracked, and acted upon.*

16.1 Toward Physically Grounded AI Systems

Recent research directions have emphasized the need for AI systems to better align with physical reality and stability principles. Theories such as predictive processing and free energy minimization have highlighted the importance of measurable system dynamics.

The Law-E framework contributes to this direction by introducing a runtime, measurable notion of stability grounded in system execution.

17. Call for Collaboration

Neomundi Lab is currently engaging with AI infrastructure teams, compliance and risk departments, and research partners to explore real-world implementations of runtime stability measurement.

Early collaborations focus on observability (OBS mode) and controlled governance scenarios (GOV mode).

Interested partners may request access to sandbox environments, technical documentation, and pilot integrations.

Reliability is no longer assumed. It is continuously evaluated.

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One stability law, many worlds of intelligence